

Ashutosh Gupta

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EDUCATION

Thapar Institute of Engineering and Technology

Bachelor of Engineering in Computer Engineering

Patiala

2022 – 2026

Cambridge School, Rewari

Class XII (PCM) — 81.4% | Class X — 82.2%

Rewari

2021 | 2019

EXPERIENCE

Virves Software Solutions

Remote | [Certificate](#)

AI Trainee

June 2026 – September 2026

- Built a FastAPI backend validating Aadhaar, PAN, and Driving License uploads using Tesseract-first hybrid OCR (EasyOCR fallback), OpenCV preprocessing, and explainable rule-based scoring with ACCEPTED, REJECTED, REUPLOAD_REQUIRED, and MANUAL_REVIEW outcomes.
- Implemented document-type detection, Aadhaar/PAN/DL number parsing, Aadhaar and PAN identity-field extraction, PAN holder-name matching, applicant tracking, manual-review approval/rejection APIs, and MongoDB/GridFS persistence.
- Cut redundant OCR work via staged candidates, resizing, rotation/crop deduplication, targeted field OCR, capped PAN OCR calls, and early exits; added pytest tests, synthetic benchmarks, and documentation.

TECHNICAL SKILLS

Languages: Python, C, C++, SQL, R

Frameworks & Libraries: FastAPI, Uvicorn, Pydantic, OpenCV, EasyOCR, Tesseract, PyMuPDF, Pillow, NumPy, PyMongo, Scikit-learn, spaCy, pytest, Requests

Databases: MongoDB, GridFS

Developer Tools: Git, GitHub, Docker, VS Code, Jupyter Notebook, Postman

Core Skills: OCR, Computer Vision, Image Preprocessing, Document Validation, REST API Development, Rule-based Scoring, Name Matching, API Key Authentication, Rate Limiting, Performance Optimization

Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Data Science, Data Mining, Statistics

PROJECTS

Resume Screener | Python, Scikit-learn, Gensim, PyPDF2, OpenAI GPT-3.5, Streamlit, Plotly

[Link](#)

- Built a resume-JD matcher using TF-IDF, Word2Vec, and keyword overlap scoring to generate an overall match percentage.
- Processed PDF resumes and raw job descriptions using PyPDF2-based text extraction and cleaning pipeline.
- Integrated GPT-3.5 to generate personalized resume improvement suggestions with an interactive Streamlit dashboard.

Fake News Detector | Python, Scikit-learn, FastAPI, spaCy, TF-IDF, Ensemble ML, Render

[Link](#)

- Trained and calibrated five classifiers on 24K+ labeled articles; the probability-averaging ensemble reached 98.4% accuracy on a 3.6K held-out set.
- Built an NLP pipeline with spaCy lemmatization, stopword removal, title weighting, and unigram + bigram TF-IDF features.
- Developed and deployed a FastAPI app with prediction, health, and model-info endpoints on Render.

ShieldAI Fraud Detection System | Python, Flask, Scikit-learn, Random Forest, SQLite, JavaScript

[Link](#)

- Trained a Random Forest model on 10,000 transactions with behavioral features, achieving 97%+ accuracy and 0.99 ROC-AUC.
- Handled class imbalance using StandardScaler pipeline and class-weight balancing for fraud detection.
- Built a Flask REST API with SQLite persistence and a JavaScript dashboard for live KPIs and risk alerts.

CERTIFICATIONS

Introduction to Python | Coding Ninjas

[Link](#)

Data Structures & Algorithms in Python | Coding Ninjas

[Link](#)

Natural Language Processing Specialization | Coursera

[Link](#)

20+ Guided Projects | Coursera